

10. Dry winds blowing from north-east and east to west of Sahara desert are known as :

- (a) Bora (b) Sirocco
(c) Harmattan (d) Mistral

Uttarakhand P.C.S. (Pre) 2024

Ans. (c)

In the Sahara Desert, dry winds blowing from the northeast and east to the west are known as Harmattan winds. These are cool, dry winds for their invigorating dryness they are also referred to as the "Doctor Wind".

The Forests

*Forests are not only natural resources but are an essential part of our environment. They play a vital role in balancing the ecosystem.

*Bhutan is one of the smallest countries in the world, but its commitment to conservation is bigger than most. Conservation of the environment is one of the four pillars of Bhutan's Gross National Happiness Philosophy. As mandated in its constitution, Bhutan preserves (at all times) 60 percent of its land under forest cover. *According to the National Forest Inventory Report, 2023 of Bhutan, the total forest area is estimated to be 69.71% (2.68 million ha.) of the total land area while 30.29% (1.16 million ha.) of the total land area is estimated to be non-forest area in 2022.

*According to ISFR, 2023, the top 10 countries in the world with the largest forest area in the year 2020 are as follows-

Forest area for top ten countries in 2020			
S. No.	Country	Forest area (000 ha)	% world forest area
1	Russian Federation	8,15,312	20
2	Brazil	4,96,620	12
3	Canada	3,46,928	9
4	USA	3,09,795	8
5	China	2,19,978	5
6	Australia	1,34,005	3
7	Democratic Republic of the Congo	1,26,155	3
8	Indonesia	92,133	2
9	Peru	72,330	2
10	India	72,160	2
Source - ISFR : 2023			

Top 10 Countries for Average Annual Net Gain in Forest Area			
Ranking	Country	Forest Area	
		1000 ha/yr	Percentage
1	China	1937	0.93
2	Australia	446	0.34
3	India	266	0.38
4	Chile	149	0.85
5	Vietnam	126	0.90
6	Turkey	114	0.53
7	USA	108	0.03
8	France	83	0.50
9	Italy	54	0.58
10	Romania	41	0.62

• **Tropical Evergreen Forests** – They are found on both the sides of the equator between 10°N and 10°S latitudes. The equatorial region remains hot throughout the year and as a result, there is a lot of rainfall throughout the year. Hence, the equatorial region is covered with evergreen forests. These forests are found in abundance in three diverse regions of the world – (1) **Amazon Basin** – The region includes Brazil, Bolivia, Peru, Ecuador, Colombia, Venezuela, French Guiana, Guyana and Suriname. (2) **Congo Basin** – The region includes the Republic of Congo, Cameroon, Central African Republic, Gabon, Equatorial Guinea and Democratic Republic of Congo. (3) **Borneo- Mekong - Southeast Asia Basin** - The region is composed of two sub-regions, the Island of Borneo and the Mekong River.

*The rain forests of Amazon Basin are the largest forest ecosystem in the world. This forest occupies at least 60% of the area of South America. If the tropical evergreen forest is cut down it does not regrow at the same pace as that of tropical deciduous forest. The reason is that excessive rain makes the soil of the tropical evergreen forests poor in nutrients. * Maximum plant species are found in the evergreen rainforests. Hardwood trees like – **Rosewood, Aabnoos, Mahogany** etc. are found in these forests.

• **Tropical Deciduous Forest** – They are also known as monsoonal forests. They are mostly found in India, northern Australia, Indonesia the Central America. There are two distinctive seasons (moist and dry) in these forests. India and its nearby regions experience three distinctive seasons thus even the forests of the area experience seasonal changes.

The main flora of these forests includes **Sal, Teak, Neem** and **Rosewood**. Apart from these other trees, herbs and shrubs are also found.

- **Temperate Evergreen Forests** – They are found in the coastal areas of mid-latitudes. They are normally found on the eastern sides of the continent such as southeast America, South China and southeastern Brazil. Temperate evergreen forests are dominated by specific species of trees. The trees have broad leaves and hardwood for example – **Oak, Loral, Magnolia**, and **Eucalyptus** etc.

- **Mediterranean Forest** – These forests are found in the mid-latitudes on the western and southwestern parts of the continent. They are located in regions receiving winter rainfall. Mediterranean forests are mostly found in the regions of Africa, Europe and Asia that lie in proximity to the Mediterranean Sea. These forests are also found in California and southwestern Australia. The main trees of the region are **Cork, Oak, Olives, Plumps, Chestnut, Pine** etc. The Mediterranean region is famous for the production of Citrus fruits like Grapes, Lemons, Oranges, Olives, Pears. Pomegranate, etc.

- **Temperate Coniferous Forests** – Also known as **Taiga Forest**. They are found between 50° to 60° North latitude. To their north they have **tundra** and to their south, they have **deciduous forests**. They are found in regions of North America, Eurasia and sub-polar regions. They are found in Canada, Europe, Asia and the USA. Temperate coniferous forests cover the largest area in the world. Important flora of the region includes – **Deodar, Fir, Hemlock, Spruce**, etc. In India, these forests are found in high Himalayan regions.

- According to data from the World Food and Agriculture Statistical Yearbook, 2024 (Data - 2022). U.S.A. is the leading producer of wood pulp in the world (47789 million tonnes) followed by Brazil and China. *FAO has defined softwood as coniferous. Conifer is a softwood tree found in abundance in the Baltic Region (Scandinavia and Russia), North America and China. * In the year 2022, India was the leading producer of wood Fuel in the world with a production quantity of 298970 thousand cubic meter, followed by China (151258 thousand cubic meters) and Brazil (133217 thousand cubic meter).

1. **Selva forests are characterized by :**

- (a) Broad leaf evergreen
- (b) Broad leaf deciduous

- (c) Coniferous evergreen
- (d) Coniferous deciduous

U.P. B.E.O. (Pre) 2019

Ans. (a)

Forests found in the Amazon River Basin in Brazil are called Selva or Selva's forests. These forests are located in the equatorial region. Due to high temperatures and above-average rainfall (200 cm) throughout the year, dense and broad leaf evergreen forests are found here. Thus, option (a) is the correct answer.

2. **A geographic area with an altitude of 400m has following characteristics:**

Months	J	F	M	A	M	J	J	A	S	O	N	D
Average maximum temp. °C	31	31	31	31	30	30	29	28	29	29	30	31
Average minimum temp. °C	21	21	21	21	21	21	20	20	20	20	20	20
Rainfall (mm)	51	85	188	158	139	121	134	168	185	221	198	86

If this geographic area were to have a natural forest, which one of the following would it most likely be?

- (a) Moist temperature Coniferous forest
- (b) Mountain Subtropical forest
- (c) Temperate forest
- (d) Tropical rain forest

I.A.S. (Pre) 2010

Ans. (d)

The annual data of temperature and rain fall show that the above features belong to the tropical rain forest.

3. **Which one among the following covers the highest percentage of forest area in the world?**

- (a) Temperate Coniferous forests
- (b) Temperate Deciduous forests
- (c) Tropical Monsoon forests
- (d) Tropical Rain forests

I.A.S. (Pre) 2003

Ans. (a)

Temperate coniferous forests cover the highest percentage of forest area in the world. Temperate coniferous forest is mainly found in high latitudes.

4. Which of the following countries has a constitutional provision for maintaining 60% of its geographical area under forest?

(a) Maldives (b) Nepal
(c) Bhutan (d) Afghanistan

U.P.P.C.S. (Mains) 2016

Ans. (c)

Bhutan has made constitutional provision for maintaining 60% of its total geographical area under forest. Presently (in 2022), 69.71 percent of Bhutan's total area is under forest cover.

5. The evergreen rainforests are found in :

(a) Australia (b) Brazil
(c) Canada (d) France

Uttarakhand P.C.S. (Pre) 2002

Ans. (b)

Tropical evergreen forests grow in the regions near the Equator (10°N-10°S) and close to the tropics. These regions are hot and receive heavy rainfall throughout the year. As there is no particular dry season, the trees do not shed their leaves altogether. This is the reason they are called evergreen. Three major regions of evergreen forest in the world are- **Amazon River Basin**- Brazil, Bolivia, Peru, Ecuador, Colombia, Venezuela, French Guiana, Guyana and Suriname. **Congo Basin**- Democratic Republic of the Congo, Cameroon, Central African Republic, Gabon, Equatorial Guinea, Republic of the Congo.

Borneo-Mekong-Southeast Asia basin- The region is composed of two sub-regions the Island of Borneo and the Mekong River. Therefore option (b) is correct.

6. Which type of forests are found near the equator –

(a) Deciduous forests
(b) Coniferous forest
(c) Meadows forest
(d) Tropical forest

I.A.S. (Pre) 2007

Ans. (d)

See the explanation of the above question.

7. Dense-forests on the earth are mostly found –

(a) Nearby Equator
(b) Nearby tropic of Cancer
(c) Nearby tropic of Capricorn
(d) Nearby Poles

41st B.P.S.C. (Pre) 1996

Ans. (a)

See the explanation of the above question.

8. Which one of the following is noted for extensive tropical rain forests?

(a) Congo Basin
(b) Ganga Basin
(c) Hwang Ho Basin
(d) Murray- Darling Basin

U.P.P.C.S. (Pre) 1996

Ans. (a)

See the explanation of the above question.

9. Taiga forests are characteristics of –

(a) Equatorial region
(b) Tropical region
(c) Sub-tropical region
(d) Temperate region

U.P.P.C.S. (Mains) 2016

Ans. (d)

Taiga forests are characteristic of temperate region. The Taiga biome is also known as coniferous forest or boreal forest. It is the northernmost biome among the temperate biomes. It extends over North America and Eurasia to the sub-polar climate regions. Summer, in this region, is comparatively shorter.

10. The world's largest and most dense forests are in-

(a) India (b) Tanzania
(c) Brazil (d) Canada

U.P. Lower Sub. (Pre) 2004

Ans. (c)

The Amazon River Basin is the largest rainforest on Earth. The basin covers 60 percent of the South American continent.

11. Which country has the highest percentage of its geographical area under forests?

(a) China (b) India
(c) Indonesia (d) Japan

U.P. U.D.A./L.D.A. (Spl) (Pre) 2010

Ans. (d)

According to the World Bank Report 2022, forest area (% of land area) of the given countries is as follows :

Japan	-	68.4%
Indonesia	-	48.0%
India	-	24.4%
China	-	23.8%

12. The coniferous forests are not found in:

(a) Amazonia (b) Scandinavia
(c) Canada (d) Finland

U.P.P.C.S.(Pre) 2001

Ans. (a)

Coniferous forests are found between 50° N and 60°N latitudes lying between the tundra to the north and the deciduous forest to the south. These forests are found in Canada, Europe, Asia, and the United States. So it is clear that coniferous forests are not found in Amazonia (Brazil). Rather, tropical evergreen forests are found in Amazonia (Brazil).

13. Coniferous forests are mainly found in –

- (a) Tropical region
- (b) Temperate region
- (c) Coastal region
- (d) None of above

Uttarakhand Lower (Sub.) (Pre) 2010

Ans. (b)

Coniferous forests are mainly found in the temperate regions of the world.

14. In which of the following countries of Africa, dense, hot and humid forest are found?

- (a) Ivory Coast
- (b) Kenya
- (c) Zimbabwe
- (d) South Africa

U.P.P.C.S. (Pre) 1998

Ans. (a)

Among the countries given in the options, hot, humid and dense forests are found in Ivory Coast. It is situated on the coast of the Gulf of Guinea.

15. Assertion (A) : Unlike temperate forests, the tropical rain forests, if cleared, can yield productive farmland that can support intensive agriculture for several years even without chemical fertilizers.

Reason (R) : The primary productivity of the tropical rain forest is very high compared to that of temperate forests.

Code :

- (a) Both (A) and (R) are true individually true, and (R) is the correct explanation of (A)
- (b) Both (A) and (R) are individually true, but (R) is not the correct explanation of (A).
- (c) (A) is true, but (R) is false.
- (d) (A) is false, but (R) is true.

I.A.S. (Pre) 2003

Ans. (d)

Instead of tropical rain forests, if temperate forests are cleared, they can yield productive farmland. Land for agriculture is being rapidly developed in mid-latitude areas by clearing these forests. So, Assertion (A) is wrong. Primary productivity of tropical rainforests (2200 grams biomass per square meter/year) is greater than that of all other types of forests. It is also greater than that of temperate forests (1320 grams of biomass/square meter/year). Thus reason (R) is true.

16. If tropical rain forest is removed, it does not regenerate quickly as compared to a tropical deciduous forest. This is because

- (a) The soil of rain forest is deficient in nutrients
- (b) Propagules of the trees in a rain forest have poor viability
- (c) The rain forest species are slow-growing
- (d) Exotic species invade the fertile soil of rain forest

I.A.S. (Pre) 2011

Ans. (a)

If the tropical rain forest is removed, it does not regenerate quickly compared to a tropical deciduous forest. The main reason is the deficiency of nutrients in the soil of the rain forest.

17. Consider the following statements :

Statement-I

The soil in tropical rain forests is rich in nutrients.

Statement-II

The high temperature and moisture of tropical rain forests cause dead organic matter in the soil to decompose quickly.

Which one of the following is correct in respect of the above statements?

- (a) Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I
- (b) Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-I
- (c) Statement-I is correct but Statement-II is incorrect.
- (d) Statement-I is incorrect but Statement-II is correct.

I.A.S. (Pre) 2023

Ans. (d)

Tropical rainforests have extremely poor soil, in contrast to the common misconception that because of the abundance of plants, warmth and moisture, the soil must be quite rich. Hence statement-I is false. Notably, tropical rainforests have higher temperatures and more moisture than other regions, which accelerates the decomposition of dead organic matter in the soil and releases nutrients more rapidly.

18. Match list I with list II and select the correct answer:

List-I (Timber)	List-II (Country)
A. Cedar	1. Myanmar
B. Douglas Fir	2. Canada
C. Mahogany	3. Mexico
D. Teak	4. Honduras

Code :

	A	B	C	D
(a)	3	2	1	4
(b)	3	2	4	1
(c)	2	3	4	1
(d)	2	3	1	4

I.A.S. (Pre) 1999

Ans. (c)

The correctly matched timbers with their countries are as follows:

Timber	Country
Cedar	– Canada
Douglas Fir	– Mexico
Mahogany	– Honduras
Teak	– Myanmar

19. Match List-I with List-II and choose the correct answer from the codes given below the lists –

List-I	List-II
A. Monsoon Forest	1. Plum and Olive
B. Equatorial Forest	2. Pine and Spruce
C. Mediterranean Forest	3. Teak and Sal
D. Coniferous Forest	4. Mahogany and Rosewood

Code :

	A	B	C	D
(a)	3	4	1	2
(b)	2	3	4	1
(c)	4	2	3	1
(d)	1	3	2	4

U.P.R.O./A.R.O. (Mains) 2013

Ans. (a)

The correct match of List-I with List-II is as follows:

Monsoon Forest	– Teak and Sal
Equatorial Forest	– Mahogany and Rosewood
Mediterranean Forest	– Plum and Olive
Coniferous Forest	– Pine and Spruce

20. Which country of the world is the largest producer and exporter of soft timber and wood pulp?

- (a) U.S.A. (b) Norway
(c) Sweden (d) Canada

45th B.P.S.C. (Pre) 2001

Ans. (d)

During the period this question was asked, Canada was the leading country for the production and export of softwood and wood pulp. According to the World Food and Agriculture Statistical Yearbook-2024 (Data-2022) United States is the top wood pulp producing country with 47789 million Tonnes in 2022. Brazil and China were in second and third place respectively.

21. Which one of the following countries is the largest producer of fuelwood in the world ?

- (a) Indonesia (b) Russia
(c) India (d) China

I.A.S. (Pre) 1997

Ans. (c)

According to data of FAO World Food and Agriculture Statistical Yearbook 2024 (Data- 2022) wood fuel production of the above countries is as follows- India (First), China (Second) and Brazil (Third).

22. Dalbergia species is associated with which one of the following ?

- (a) Cashew nut (b) Coffee
(c) Tea (d) Rosewood

I.A.S. (Pre) 2007

Ans. (d)

Dalbergia species is a large group that has 100-600 species. Dalbergia is a large genus of small to medium-sized trees, these trees are mainly found in North and South America, Africa, Madagascar and the tropical region of South Asia. Some important species of Dalbergia are Sheesham (Rosewood), Kingwood, tulipwood and money bush etc. The smell of sesame or Rosewood is like that of a rose, so it is called Rosewood.

23. Which of the following leaf modifications occur(s) in the desert areas to inhibit water loss?

1. Hard and waxy leaves
2. Tiny leaves
3. Thorns instead of leaves

Select the correct answer using the code given below:

- (a) 2 and 3 only (b) 2 only
(c) 3 only (d) 1, 2 and 3

I.A.S. (Pre) 2018

Ans. (d)

Desert vegetation has the ability to tolerate high temperatures and dry weather. Such vegetation is called erophyte vegetation. Here, the plants take several measures to adjust to the hot and dry climate. The roots of desert plants are long so that they can penetrate more deeply and obtain moisture. The bark of the stem is thick. Leaves are either absent or if they are present, they are small waxy and thick, with small spines on them and closed pores to prevent evapotranspiration. Hence, option (d) is the correct answer.

24. Match List- I with List- II and select the correct answer using the code given below the lists :

List- I (Vegetation Type)	List- II (Region)
A. Maquis	1. California
B. Fynbos	2. Mediterranean Sea Coastal Region
C. Chaparral	3. South Africa
D. Matorral	4. Chile

Code :

	A	B	C	D
(a)	2	3	1	4
(b)	4	3	2	1
(c)	3	4	1	2
(d)	4	2	3	1

U.P.P.C.S. (Pre) 2019

Ans. (a)

The correct match of List-I with List-II is as follows :

Maquis	-	Mediterranean Sea Coastal Region
Fynbos	-	South Africa
Chaparral	-	California
Matorral	-	Chile

As, per question the correct answer is option (a).

25. Match List- I with List- II and select the correct answer using the code given below the lists :

List- I (Natural Vegetation)	List- II (Region)
A. Epiphytes	1. Mediterranean
B. Acacia	2. Equatorial
C. Baobab	3. Sahara
D. Cedars	4. Savanna

Code :

	A	B	C	D
(a)	2	3	4	1
(b)	2	3	1	4
(c)	2	4	3	1
(d)	2	4	1	3

U.P.P.C.S. (Pre) 2019

Ans. (a)

The correct match of List-I with List-II is as follows :

List- I (Natural Vegetation)	List- II (Region)
Epiphytes	- Equatorial
Acacia	- Sahara
Baobab	- Savanna
Cedars	- Mediterranean

The correct answer is option (a).

26. "Leaf litter decomposes faster than in any other biome and as a result the soil surface is often almost bare. Apart from trees, the vegetation is largely composed of plant forms that reach up into the canopy vicariously, by climbing the trees or growing as epiphytes, rooted on the upper branches of trees." This is the most likely description of

- (a) coniferous forest
- (b) dry deciduous forest
- (c) mangrove forest
- (d) tropical rain forest

I.A.S. (Pre) 2021

Ans. (d)

Due to the high temperatures and increased rainfall in the tropical region throughout the year, there is a diversity of trees in the forests of this area. Here epiphytes are also found, and the fallen leaves rapidly rot and decompose.

27. In which forest zone of the world highest deforestation has been recorded?

- (a) Coniferous
- (b) Tropical
- (c) Savanna Deciduous
- (d) Tidal

Raj. P.C.S. (Pre) 2024

Ans. (b)

According to The State of the World's Forests report - 2024, in the year 2020, about 4.1 billion hectares (31%) of the world's land surface was covered by forests with the largest share in the tropical region. The highest deforestation has also been recorded in tropical areas.

28. Consider the following trees :

- 1. Jackfruit (*Artocarpus heterophyllus*)
- 2. Mahua (*Madhuca indica*)
- 3. Teak (*Tectona grandis*)

How many of the above are deciduous trees?

- (a) Only one
- (b) Only two